



Unresolved issues with efgartigimod in CIDP, and broader concerns for future trial design

The U.S. Food and Drug Administration's (FDA) approval of efgartigimod, an FcRn antagonist, for the treatment of chronic inflammatory demyelinating polyneuropathy (CIDP) was based on the positive results of the ADHERE trial, which demonstrated that efgartigimod significantly outperformed placebo in a double-blind withdrawal study [1]. However, a recent case series by Levine and Muley reports a concerning adverse outcome not highlighted in the trial [2]. This new finding underscores the need for heightened safety vigilance and raises important questions about both the trial design and the broader implications for clinical research in CIDP.

In the case series, four out of nine patients, all treated post-approval, experienced severe disease worsening after transitioning from intravenous immunoglobulin (IVIg) to efgartigimod. INCAT scores increased by 2 to 5 points within two weeks of initiating treatment. Two patients required hospitalization, and none had returned to baseline despite receiving rescue therapy at the time of reporting. The final dose of IVIg occurred 12 to 42 days before the initiation of efgartigimod therapy.

The timeline aligns with a high dropout rate among patients who stopped immunoglobulin therapy in the ADHERE trial. The protocol required a run-in phase, during which patients stopped prior treatment and were monitored for disease worsening over 12 weeks. Over 90 % of cases worsened, allowing progression to stage A, where subjects started open-label efgartigimod. Only patients who improved after four doses advanced to a randomized double-blind withdrawal (stage B), where they continued efgartigimod or switched to placebo.

By the end of stage A, 53 of 165 subjects who had been on immunoglobulin therapy (32 %) dropped out, some within weeks of starting treatment. In contrast, patients on prior corticosteroids or no prior therapy had lower dropout rates (13 % and 9 %, respectively).

It is plausible that some ADHERE participants experienced severe deterioration, similar to patients in the Levine and Muley case series, but the trial data does not provide clarity. It remains difficult to determine whether these 4 patients represent an isolated event or indicate broader underreporting. Investigators did not design the ADHERE to capture outcomes for patients after they dropped out, leaving unanswered questions about responses to rescue interventions, the severity of worsening compared to typical IVIg relapses, or the occurrence of serious outcomes requiring hospitalization. While the adverse event results note that 17 patients (5 % of the total study) experienced "CIDP worsening" during stage A, we do not know what prior treatments these patients were taking. If all 17 had been on immunoglobulin, it would suggest approximately 10 % of the subgroup worsened (17/165). However, this calculation does not fully explain the high dropout rate.

The ADHERE report also lists reasons for patient withdrawals but groups them into broad categories such as "participant withdrawal,"

"insufficient efficacy," and "adverse events," limiting interpretability. This lack of detail creates gaps in understanding the risk that newly treated patients might face. While the report acknowledges the high dropout rate among immunoglobulin-treated patients and the need to find the "optimal approach" to transition therapy, the authors downplay concerns, adding that dropouts might have responded differently if they had completed their dosing schedules.

CIDP criteria have come full circle

Since the Ad Hoc AAN subcommittee first published CIDP research criteria in 1991, guidelines have evolved significantly [3]. Stated purposes of their work were to "engender" discussion about CIDP, to recognize the complexity of the condition, and to prioritize specificity over sensitivity in the context of clinical research. They defined disease attributes that were not meant to encompass all cases. However, these criteria eventually proved too restrictive for clinical practice, leading to the EFNS/PNS guidelines in 2005, which aimed to improve sensitivity and to make it easier for practitioners to treat patients [4]. Ironically, the ADHERE trial may have overlooked these foundational concepts by using, as the title implies, "management guidelines" for research participant selection, potentially leading to over-diagnosis and affecting the study's rigor.

Estimates of misdiagnosis rates for CIDP in the community are problematic, ranging from 50 % [5] to near 70 % [6] depending on definitions. To reduce misdiagnosis, ADHERE researchers relied on a committee of independent expert adjudicators and used the updated 2010 EFNS/PNS guidelines as study criteria [7]. Two adjudicators reviewed each case, and inclusion required two 'yes' votes. If adjudicators disagreed, a committee chairperson acted as the tiebreaker.

Poor concordance raises questions about diagnosis

The novel study design employed by the ADHERE investigators addressed limitations in past CIDP trials, which lacked enrichment strategies and adjudication procedures. Some trials faltered due to a combination of patients without active disease and the confounding effects of prior treatments [8]. Selecting the appropriately diagnosed patients is critical to focus on desired effects and minimize "noisy" outcomes from misdiagnosis or disease remission.

Two early abstracts, after screening 100 and then 200 cases, reported a discordance rate of about 50 % between the first two adjudicators [9]. This high discordance rate is a red flag, suggesting potential random guessing or significant differences in how adjudicators interpreted ambiguous cases. Statistically, the inter-rater reliability for the case

review process points to a relatively low kappa value, indicating that the approach may not have effectively limited some number of misdiagnosed cases. The low agreement rate points to a lack of consensus among adjudicators, undermining the goal of accurately identifying CIPD patients for valid inclusion in the study.

The adjudication challenges go beyond the decision to use guidelines meant for clinical practice. ADHERE tasked adjudicators with evaluating documentation from multiple countries, some based on clinic notes not created directly for trial enrollment. Like judges reviewing circumstantial evidence, adjudicators had to interpret case material, including some borderline findings—such as two isolated conduction blocks with no other abnormalities—that might meet EFNS/PNS guidelines but not the 1991 research criteria. In such an instance, it is hard to know how any given adjudicator would decide a confusing case. Research on decision-making suggests evaluators bring personal biases to similar processes, influenced by factors such as strictness with rules, responses to grammatical context, or even mood at the time of review [10]. Given the ambiguity in both the guidelines and the data, it is unsurprising that agreement among adjudicators was somewhat random. Notably, two adjudicators could enroll a subject simply by guessing the same way on a hard case.

Concern over placebo responses and incentives to respond to treatment

In a systematic review of earlier CIPD trials, Lewis and colleagues concluded that “placebo is an important variable in the design of CIPD trials” [11]. However, the paper never explicitly stated that misdiagnosis is a major contributor to placebo responses. ADHERE, as a withdrawal study, was prone to such effects. Beyond adjudication, other observations raise concern about potential misdiagnosis. For example, a high percentage of patients in the placebo group in stage B never relapsed, despite rapid relapses during the lead-in phase. While the ADHERE report mentions potential carryover effects of efgartigimod, similar phenomena are known in other CIPD trials [8,11]. Additionally, subjects who had previously received immunoglobulin responded weeks faster to efgartigimod in stage A than those on prednisone or no prior therapy, suggesting that learned expectations, rather than immune effects alone, may have influenced the observed responses. Prior immunoglobulin patients who relapsed in stage B also did so more rapidly than those in other cohorts.

In clinical practice, patients misdiagnosed with CIPD often report positive responses to treatments [5]. CIPD specialists sometimes encounter misdiagnosed patients who have remained on treatment for years. These individuals, essentially proven placebo responders, pose unique challenges for clinical trialists, especially when incentivized to show positive treatment responses.

Using open-label extensions (OLE) in clinical trials can attract participants who hope to transition to a new treatment but may also create incentives to respond to therapy. For example, it is hard to simply accept that 90 % of ADHERE patients relapsed during the run-in phase, compared to just 40 % in other trials when prior treatments were discontinued [8]. This is too large a difference to ignore a potential effect of incentives, in addition to blinding and patient selection. In addition, the OLE design may have inflated the observed response rate, as misdiagnosed participants also had to show clinical improvement during stage A to gain entry to the OLE. This potentially introduced a well-known response bias [12]. The open-label design of stage A, without a placebo control, removed a key safeguard against placebo responses. Combined with the potentially liberal criteria used in the adjudication process, some findings in the trial likely reflect placebo responses rather than efgartigimod's efficacy in CIPD.

Furthermore, by the end of stage A, the design could have inadvertently enriched stage B with misdiagnosed patients motivated to show improvement. This contrasts with correctly diagnosed patients, who may have experienced worsening or severe relapses and dropped out.

The outcome measures used in ADHERE—such as I-RODS, INCAT, and grip strength—are subjective and prone to participant effort, making them insufficient to address this potential confounding factor.

It follows that throughout stage B, the risk of unblinding required an additional guardrail, especially considering misdiagnosed subjects prone to placebo responses. In a typical head-to-head treatment trial, blinding helps mitigate this issue, but in ADHERE, one arm transitioned directly from active treatment to placebo, while the other did not. If participants sensed a change from the known therapy to a placebo, their expectations could influence their responses—particularly with a drug that causes injection site irritation. Any sense of unblinding could skew relapse rates, with the placebo group experiencing higher rates and the treatment group lower rates. The ADHERE report does not address unblinding, nor does it mention whether they queried participants about their assumptions regarding treatment assignment.

Final thoughts and the opportunity in the existing data

The report by Levine and Muley highlights critical safety concerns about transitioning patients from immunoglobulin therapy to efgartigimod, potentially shedding light on the dropout patterns in the ADHERE trial. We cannot conclude efgartigimod is ineffective or dangerous for some patients, but these insights call for broader discussion on study designs and require a reevaluation of the trial's findings.

First, an intensive and transparent phase 4 safety monitoring program is crucial, with systematic reporting of severe relapses. If the incidence of deterioration proves high, a warning should be issued to inform patients and clinicians of the risks. Reassessing the outcomes of patients who dropped out of the ADHERE trial and required rescue therapy could also provide valuable clarity. Second, while the reported trial results seem encouraging, further data and detailed case reporting are still needed to confirm long-term safety and efficacy, particularly for newly diagnosed or untreated patients, where documenting improvement is more straightforward than for patients transitioning treatments who remain stable. Third, a comprehensive sensitivity analysis of the ADHERE data is necessary to explore critical likelihoods that the study did not fully account for, such as misdiagnosis rates, placebo effects, response biases in open-label stages, unaccounted dropouts, and unblinding risks. This analysis could help exclude a gross overestimation of the drug's safety and efficacy. Finally, the variability in adjudicated diagnoses offers an opportunity to refine CIPD diagnostic criteria and address methodological flaws that have led to noisy outcomes in several CIPD trials. A post-hoc analysis examining recognizable subsets—like dropouts needing rescue therapy, placebo non-relapsers, or newly diagnosed responders—who had vastly different outcomes despite following the same protocol, is needed to improve diagnostic precision and strengthen confidence in efgartigimod's role in CIPD. Addressing these issues will only ensure that both patients and clinicians benefit from a clearer, more accurate understanding of promising therapies.

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